Week 7 question sheet

Problem sheet

Instructions

Attempt all ten questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator or software is needed. Give exact values unless a question says otherwise.

For every equality-constrained problem, state the constraint encoding before interpreting a multiplier. For every inequality-constrained maximisation problem, use

$$g_j(\mathbf{x}) \geq 0, \qquad \mathcal{L} = f + \sum_j \lambda_j g_j + \sum_k \mu_k h_k, \qquad \lambda_j \geq 0.$$

Write stationarity, primal feasibility, dual feasibility and complementary slackness. State the parameter space before maximising a likelihood.

Use hints.md only after a serious attempt. Do not consult solutions.md until you have completed or paused every part.

Question 1 — equality-constrained allocation (`M10`; 12 marks; 26 minutes)

For $x > 0$ and $y > 0$, consider

$$\max_{x,y} \{\log x + 2 \log y\} \quad \text{subject to} \quad x + 2y = 12.$$

`W07-Q01-PA` — **5 marks; 12 minutes** Encode the constraint as $12 - x - 2y = 0$. Form the Lagrangian, solve the first-order and feasibility conditions, and report the candidate (x^*, y^*) and multiplier μ^*.

`W07-Q01-PB` — **7 marks; 14 minutes** Justify why the candidate is the unique global maximiser. Then replace 12 by a resource parameter $a > 0$, derive the value function $V(a)$, and verify at $a = 12$ that the envelope derivative agrees with the multiplier under the stated encoding. Interpret the sign of the derivative.

Question 2 — a parameterised equality constraint (`M10`; 11 marks; 27 minutes)

For $a \in \mathbb{R}$, consider

$$\max_{x,y} \left\{-(x-2)^2 - (y-1)^2\right\} \quad \text{subject to} \quad x + y = a.$$

Use $h(x, y, a) = a - x - y = 0$.

`W07-Q02-PA` — **7 marks; 16 minutes** Derive $x^*(a)$, $y^*(a)$, $\mu^*(a)$ and the value function $V(a)$. State a sufficient condition for the global conclusion.

`W07-Q02-PB` — **4 marks; 11 minutes** Use the envelope theorem to find $V'(a)$ without differentiating the optimisers. Reconcile the result with direct differentiation of $V(a)$. Explain what happens to the multiplier if the same constraint is instead written $x + y - a = 0$.

Question 3 — two equality constraints (`M10`; 10 marks; 20 minutes)

`W07-Q03-PA` — **10 marks** For $a \in \mathbb{R}$, solve

$$\max_{x,y,z} -(x^2 + y^2 + z^2)$$

subject to

$$x + y + z = a, \qquad x - y = 0.$$

Encode the first constraint as $a - x - y - z = 0$ and the second as $x - y = 0$. Derive both multipliers and the value function, check the equality-constraint regularity condition, justify uniqueness, and verify the envelope derivative with respect to a. Report all results also at $a = 6$.

Question 4 — Kuhn–Tucker conditions with a capacity bound (`M11`; 12 marks; 25 minutes)

Consider

$$\max_x \; 8x - x^2 \quad \text{subject to} \quad x \geq 0, \qquad 3 - x \geq 0.$$

`W07-Q04-PA` — **5 marks; 10 minutes** Write the Lagrangian and the complete Kuhn–Tucker conditions under the course convention. Do not solve them yet.

`W07-Q04-PB` — **7 marks; 15 minutes** Solve the conditions by checking the possible interior and boundary cases. Report the maximiser and both multipliers, then give a sufficient-condition argument for the unique global conclusion.

Question 5 — two inputs and explicit corner checks (`M11`; 13 marks; 31 minutes)

Consider

$$\max_{x,y} \{6x + 4y - x^2 - y^2\}$$

subject to

$$x \geq 0, \qquad y \geq 0, \qquad 4 - x - y \geq 0.$$

`W07-Q05-PA` — **7 marks; 16 minutes** Form the Lagrangian and the complete Kuhn–Tucker conditions. Solve the branch on which $x > 0$, $y > 0$ and the resource constraint binds.

`W07-Q05-PB` — **6 marks; 15 minutes** Check the budget-slack branch and the boundary branches $x = 0$ or $y = 0$. Explain why no other branch supplies a Kuhn–Tucker point. Use curvature and the feasible set to justify the unique global maximiser.

Question 6 — diagnosing a convention error (`M11`; 6 marks; 14 minutes)

`W07-Q06-PA` — **6 marks** A student wants to solve

$$\max_x \; 4x - x^2 \quad \text{subject to} \quad 0 \leq x \leq 1.$$

The student writes only $g(x) = x - 1 \geq 0$ and $\mathcal{L} = 4x - x^2 - \lambda g(x)$, with $\lambda \geq 0$.

Identify all convention and feasibility errors. Rewrite both bounds as course-convention constraints, state the complete Kuhn–Tucker conditions, and solve for the maximiser and both multipliers.

Question 7 — terminology and estimator properties (S7; 6 marks; 14 minutes)

W07-Q07-PA — **6 marks** Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} F_\theta, \qquad \mathbb{E}_\theta[X_i] = \theta, \qquad \operatorname{Var}_\theta(X_i) = \sigma^2 < \infty,$$

where σ^2 is known. Consider

$$\hat{\theta}_n = \bar{X} + \frac{1}{n}.$$

Distinguish the parameter, estimand, estimator and observed estimate. For $n = 4$ and observed data $(1, 2, 3, 4)$, calculate the estimate. Then derive the bias, variance and mean squared error of $\hat{\theta}_n$, and prove that it is consistent for θ.

Question 8 — method of moments (S7; 6 marks; 16 minutes)

W07-Q08-PA — **6 marks** Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta), \qquad \theta > 0.$$

Using $\mathbb{E}_\theta[X_i] = \theta/2$, derive the one-moment method-of-moments estimator of θ. For the observed sample $(1, 2, 2, 3)$, calculate the estimate and check whether the fitted support contains every observation. Derive the estimator's bias and state, with a reason, whether it is consistent.

Question 9 — elementary maximum likelihood (S7; 16 marks; 37 minutes)

W07-Q09-PA — **8 marks; 17 minutes** Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \qquad p \in [0, 1].$$

For observed values $x_i \in \{0, 1\}$, write $s = \sum_{i=1}^{n} x_i$. Derive the maximum-likelihood estimator of p. Your argument must handle separately the interior case $0 < s < n$ and the boundary samples $s = 0$ and $s = n$.

W07-Q09-PB — **8 marks; 20 minutes** Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda), \qquad \lambda > 0,$$

where λ is the rate and the density is $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Derive the maximum-likelihood estimator, including the maximum check. For the observed sample

$$\left(\frac{1}{2}, 1, \frac{3}{2}, 1\right),$$

calculate the estimate. Derive the method-of-moments estimator and explain their relationship in this model.

Question 10 — cumulative retrieval (M8, M9, S6; 8 marks; 30 minutes)

Complete these three parts without consulting earlier notes.

`W07-Q10-PA` — `M8`; **3 marks; 10 minutes** Let

$$\mathbf{A} = \begin{bmatrix} -4 & 2 \\ 2 & -2 \end{bmatrix}, \qquad q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}.$$

Classify the quadratic form by definiteness using an appropriate test. State the corresponding curvature conclusion for $f(\mathbf{x}) = q(\mathbf{x}) + \mathbf{b}^{\mathsf{T}}\mathbf{x} + c$ on $\mathbb{R}^2$, where $\mathbf{b} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

`W07-Q10-PB` — `M9`; **3 marks; 10 minutes** Find and classify the global solution over $(x, y) \in \mathbb{R}^2$ of

$$\max_{x,y}\{4x + 2y - x^2 - y^2\}.$$

State both the candidate condition and the sufficiency argument.

`W07-Q10-PC` — `S6`; **2 marks; 10 minutes** Let

$$X_1, \dots, X_{36} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 9).$$

Derive the exact sampling distribution of $\bar{X}$ and calculate $\mathbb{P}(|\bar{X} - \mu| \leq 1)$. You may use $\Phi(2) = 0.9772$, where Φ is the standard normal CDF.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Use the course constraint convention and state every parameter space or sufficiency condition that affects your conclusion. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W07-ST01-PA` — **equality constraint and envelope (`M10`; 5 marks; 7 minutes)** For a resource parameter $a \in \mathbb{R}$, consider

$$\max_{x,y}\{-(x-1)^2 - (y-2)^2\} \quad \text{subject to} \quad x + y = a.$$

Encode the constraint as $a - x - y = 0$. Derive the optimisers and multiplier as functions of a, justify the global conclusion, and verify the envelope derivative at $a = 4$.

`W07-ST02-PA` — **an upper-bound corner (`M11`; 5 marks; 6 minutes)** Solve

$$\max_{x}\ 6x - x^2 \quad \text{subject to} \quad x \geq 0, \qquad 2 - x \geq 0$$

using the course Kuhn–Tucker convention. State the complete conditions and report both multipliers.

`W07-ST03-PA` — **bias, variance and consistency (`S7`; 5 marks; 6 minutes)** Let $X_1, \dots, X_n$ be iid with

$$\mathbb{E}_\theta[X_i] = \theta, \qquad \mathrm{Var}_\theta(X_i) = 4.$$

For

$$\hat{\theta}_n = \bar{X} - \frac{1}{n},$$

derive the bias, variance and mean squared error, and state whether the estimator is consistent.

`W07-ST04-PA` — Poisson maximum likelihood (`S7`; 5 marks; 6 minutes) Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda), \qquad \lambda > 0.$$

Derive the maximum-likelihood estimator of λ for a sample with a positive total count. Calculate the estimate for $(0, 1, 2, 1)$, and state the one-moment method-of-moments estimator.